Kenneth Gampol

US Citizen | Active Secret Clearance | Willing to Relocate

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Education

University of Pennsylvania **Jan 2025 – Dec 2025**
M.S. in Computer Information Technology (Computer Science), GPA: **3.8/4.0** Philadelphia, PA

University of California Merced **Aug 2021 – Dec 2024**
B.S. in Mechanical Engineering, Magna Cum Laude, GPA: **3.8/4.0** Merced, CA

Technical Skills

Languages: Python, C++ (14/17/20), Java, Rust, TypeScript, C#, SQL, Bash
AI & Machine Learning: PyTorch, TensorFlow, HuggingFace, LLM Inference, RAG, Agentic AI, OpenAI API, BERT
AI Infrastructure: VectorDBs, Prompt Guardrails, Observability, CUDA, TCP/UDP, ROS 2, Docker, Kubernetes
Frameworks & Cloud: AWS, React, FastAPI, Flask, Node.js, PySpark, Pandas, NumPy, CI/CD, Git

Experience

The Boeing Company **May 2025 – Present**
AI/ML Software Engineer Hazelwood, MO

- Designed a scalable AI pipeline leveraging Agentic workflows to automate C code generation; partnered with cross-functional teams and pitched to executive leadership, securing funding for full deployment.
- Built a C# pre-mission validation pipeline that runs pilots' planned trajectories through a simulation harness, extracts flight data, and automatically flags anomalies, cutting mission-plan review from months to hours; now extending it toward near-real-time validation and onboard dynamic mission planning.
- Engineered a BERT-based semantic verifier (PyTorch/HuggingFace) for automated model evaluation and guardrails, scoring LLM outputs against specs to ensure responsible AI alignment and strict safety compliance.
- Optimized hardware utilization and inference latency by architecting a zero-allocation C++ UDP multicast streaming library; feeds an NVIDIA Thor platform running CUDA-accelerated CV for real-time edge inference.
- Developed a static-memory C++ formation flight controller for autonomous aerial refueling under DO-178C deterministic constraints, managing directed waypoint graphs and stage/engage state machines.
- Built a TCP/UDP client-server simulation backend integrated into proprietary aerodynamic frameworks, enabling high-throughput multi-ship Monte Carlo analysis for manned-unmanned teaming (MUM-T).

The Boeing Company **May 2024 – May 2025**
Structural Stress Intern Long Beach, CA

- Built a fully automated Python optimization tool that ingests FEA load cases from HyperMesh, structures data in Pandas, and streams results via UDP to iteratively resize components to optimal margin of safety at minimum weight.
- Applied the optimizer to resize a 3-stage telescoping tube under clearance constraints, cutting excess weight and enabling rapid programmatic sizing across design changes.
- Recognized as a Top 100 Intern in the Nation by WayUp and Yello for technical contribution and organizational impact.

Sun Maid Growers of California **Dec 2023 – Dec 2024**
Automation/Systems Engineer Kingsburg, CA

- Deployed a computer vision system with GPIO-controlled air rejection to detect and remove defective packaged goods at full conveyor line speed, successfully maintaining high system throughput.
- Projected $400K in labor savings over six years and initiated scaling across all packaging lines; recognized as Top Capstone Project Finisher.

Research

University of Southern California **April 2024 – August 2024**
Research Assistant, Safety Intelligence & Autonomy Lab Los Angeles, CA

- Implemented real-time safe trajectory planning on an autonomous RC car using Hamilton-Jacobi reachability on NVIDIA Jetson; built full LiDAR-to-actuation pipeline via ROS 2 to maximize inference and control latency.

Projects

MasterMind AI – Wharton AI Hackathon (Top 10 Finalist) | ***Python, VectorDBs, RAG, GPT-4***

- Shipped a full-stack Retrieval-Augmented Generation (RAG) study assistant in 24 hours that ingests lecture PDFs and utilizes embedding-based similarity search to ground model outputs in factual context.
- Engineered a scalable prompt orchestration and caching layer that optimized system performance, successfully reducing API token usage (cost) and inference latency by ˜40%.

Autonomous RC Car | ***Python, TensorFlow, SLAM, LiDAR***

- Built an end-to-end autonomous vehicle on Raspberry Pi + LiDAR: SLAM-based localization, stop-sign perception via CNN, and on-device path planning. Led to a cold-contact research position at USC's Autonomy Lab.